Feature superposition and interference in convolutional neural networks

TER Master 1 · Sciences de Données et Systèmes Complexes

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Introduction

Context

- CNNs **not only accurate, but also efficient**.
- Important for **resource-constrained devices**.
- Architectural choices can reduce **computation and memory costs**, but also **influence how features are internally organized.**
- Understanding this influence remains a **major challenge in interpretability**.
- Feature **superposition** can be **efficient** but also create **interference**.

Presentation plan

- **Theoretical foundations** — what superposition is, how it works, and why it is linked to architectural design.
- **Experimental part** — compare different CNN block designs and analyze their internal representations.

Main question

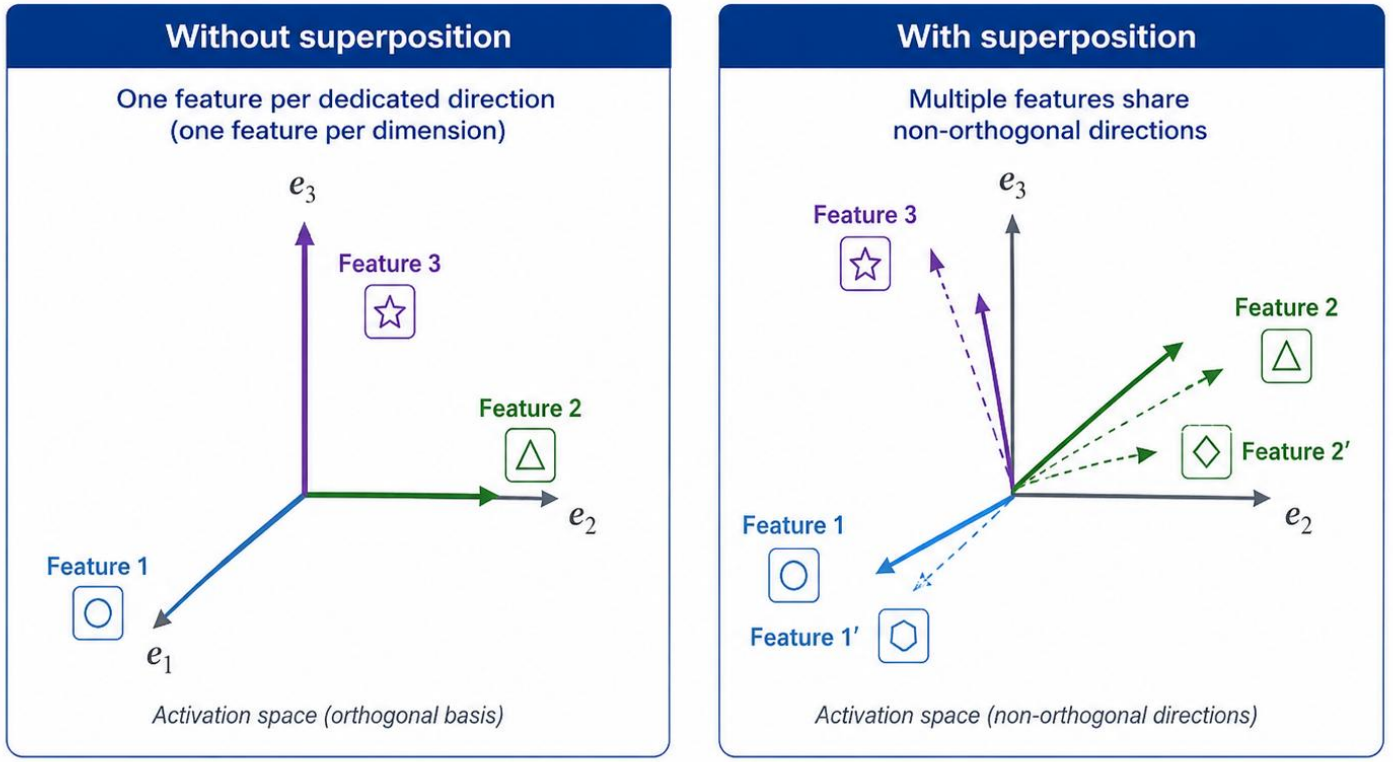
How does block design influence feature organization in CNNs,

and what does it suggest about superposition and interference?

Superposition as a structural constraint

Superposition

- Superposition means that a network **can encode more features than its dimensions would normally allow**.
- It does this **by storing multiple features in shared directions** of the activation space.
- Instead of giving each feature its own **dedicated direction**.



Sparse features can share directions because they rarely co-occur.

If shared features co-activate, interference can appear.

How it occurs and how it is useful

Useful under dimensional constraints

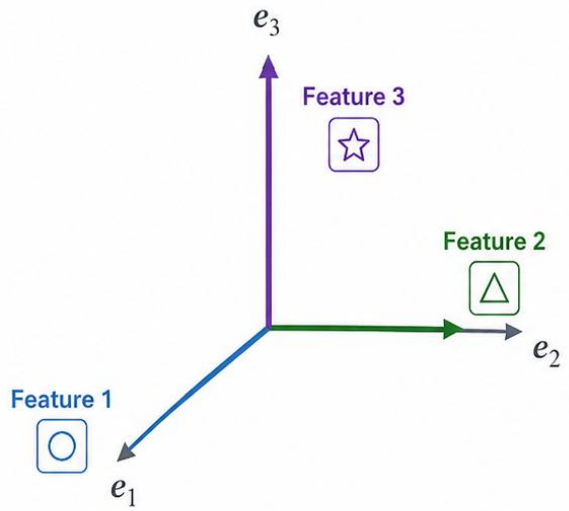
- **Efficient** when the network has a limited number of dimensions to represent features.
- **Packing more features** into the same activation space.
- Can perform better than a linear model, while using fewer dimensions.

Comes with a cost : interference

- If features sharing a direction become active together, their signals can mix.
- Can degrade representation quality.
- Makes interpretation more difficult.

Without superposition

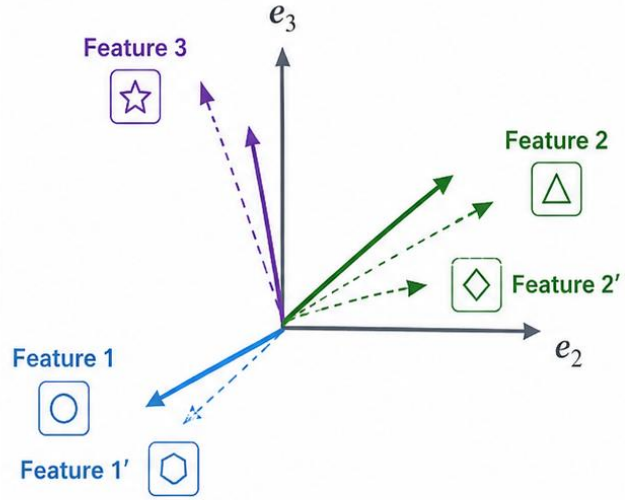
One feature per dedicated direction
(one feature per dimension)



Activation space (orthogonal basis)

With superposition

Multiple features share
non-orthogonal directions



Activation space (non-orthogonal directions)

Sparse features can share directions because they rarely co-occur.

If shared features co-activate, interference can appear.

Efficient representation, but harder interpretation.

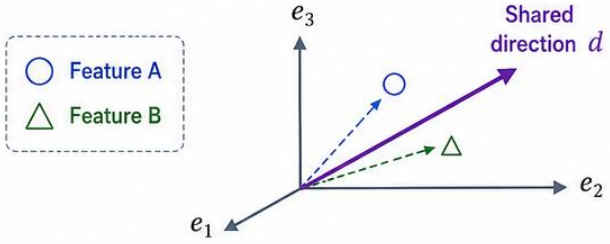
Role of sparsity in enabling superposition

Rarely co-activate

1) **Rare activation in data** (mostly at different times)



2) **Activation space:** same shared direction

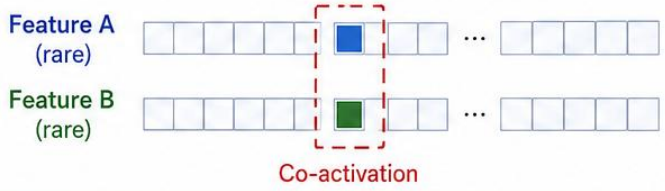


Sparse features can share a direction because they rarely co-occur.

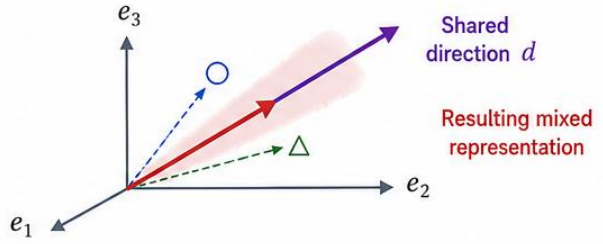
Same shared direction → little interference when activations do not overlap.

If they co-activate

1) **Co-activation example** (happens rarely)



2) **Activation space:** signals add in the same direction



If shared features co-activate, interference can appear.

The representation becomes **harder to separate** and interpret.

Sparse features

- Play key role in making superposition possible.
- Features that occur **rarely**, and especially rarely at the same time.
- The network can place sparse features in the same directions as more frequent ones.
- Since they almost never **co-occur**, they can share dimensions without interfering.
- But if rare features do co-occur, their representations can mix and create **interference**.

Phase change to superposition

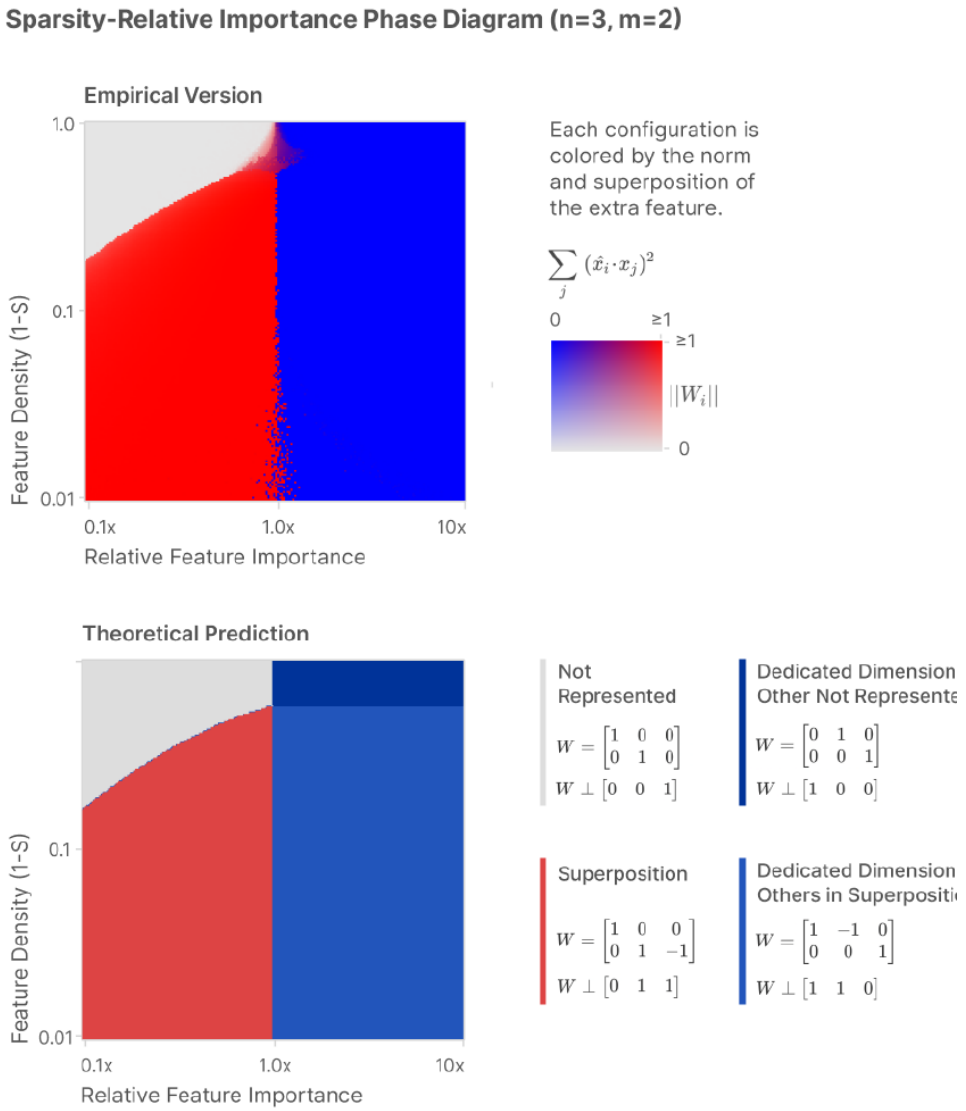


Figure: phase diagram of feature representation during training. Source: Elhage et al. (2022).

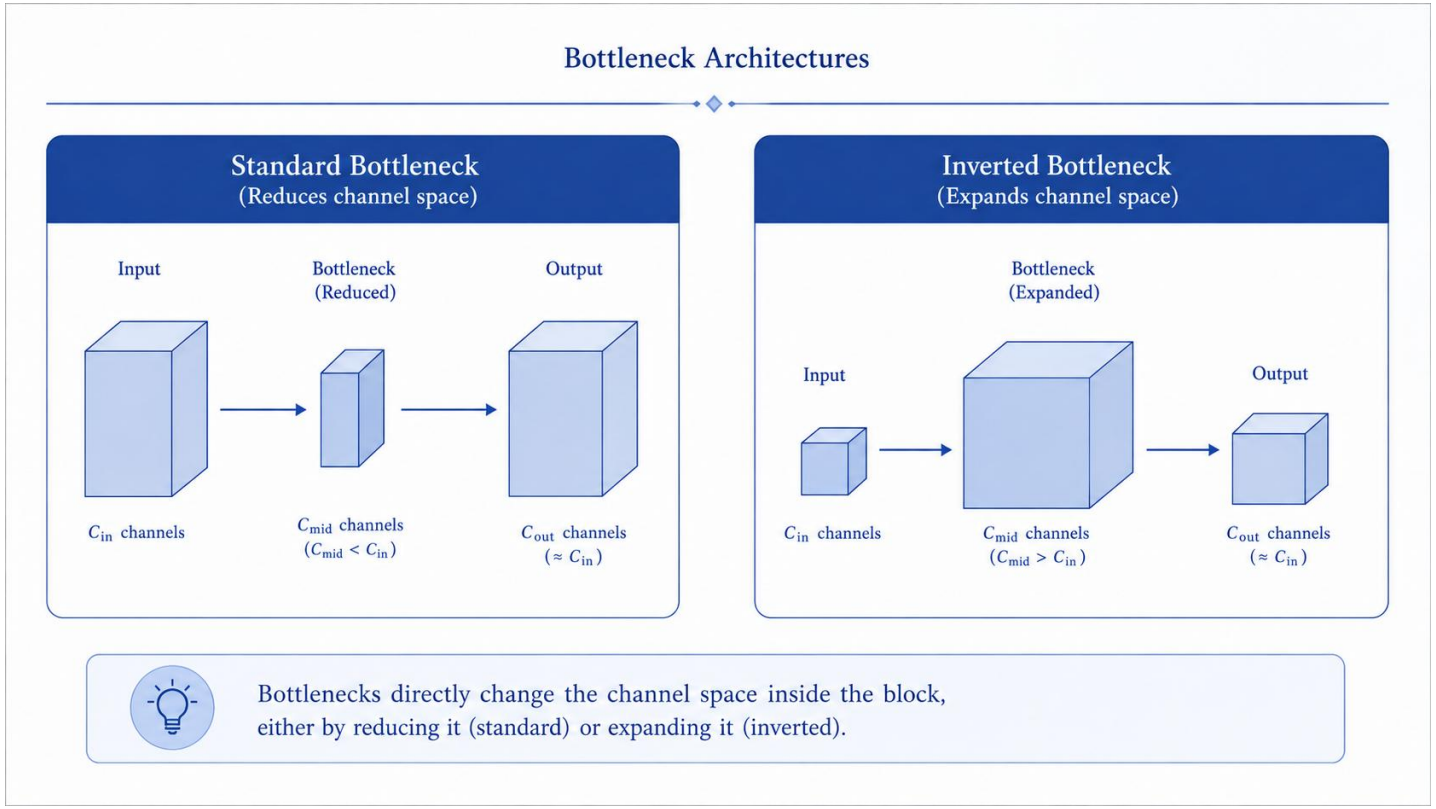
Key idea

- Superposition does **not emerge gradually.**
- Can appear through a **phase change**. Driven by **sparsity** and **relative feature importance**.
- **Relative feature importance** : the gain to prediction loss from representing a feature.
- After threshold, the model can switch from **orthogonal dimensions** to **shared directions**.
- During training, a feature can be **not represented**, represented in **superposition**, or given its own **orthogonal direction**.

Feature representation and architectural design

Why bottlenecks are relevant

- Feature representations are affected by **dimensional constraints**.
- **Bottleneck architectures** are relevant because they directly change the channel space.
- A standard bottleneck **reduces** the channel space.
- An inverted bottleneck **expands** the channel space.
- This helps study how dimensional constraints can create **representational overlap**.
- This overlap may be connected to **potential superposition**.



Experiment: context and motivation

Experimental goal

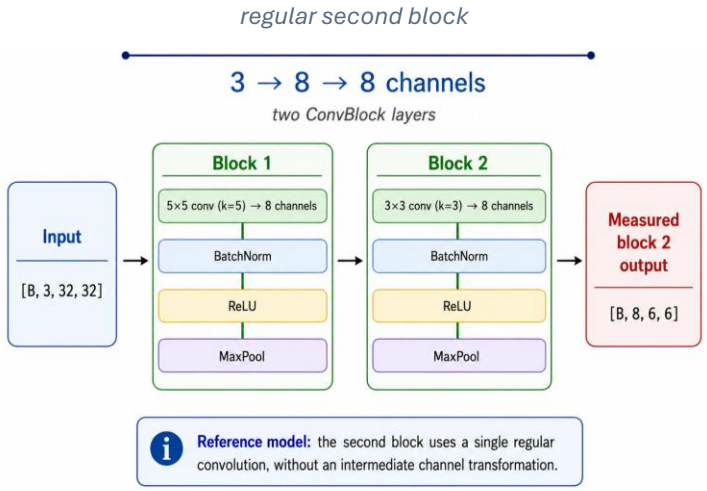
The experimental part connects the theory to a practical comparison between different CNN block designs**. The goal is not to prove superposition directly, because there is currently no reliable and direct measure for that**, but **to see if different blocks produce different geometric signatures in their representations.**

Experimental question

Do standard and inverted bottleneck blocks produce different levels of channel overlap and effective representational dimensionality ? And are these patterns consistent with superposition-like feature organization?

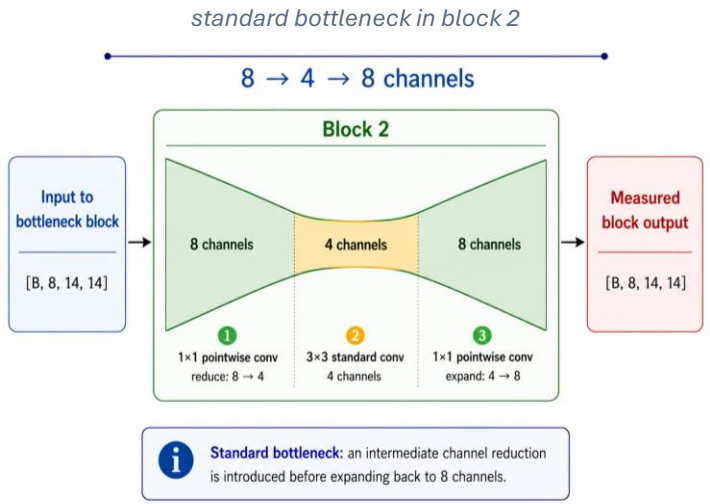
Setup: compared architectures

BaselineConvNet



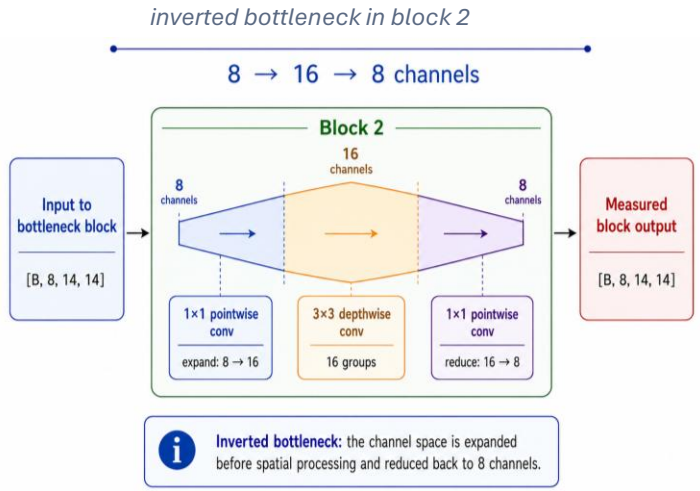
Two-block CNN
Block 2 is the studied block
Block 2 is a regular convolutional block.
Reference model

StandardBottleneck



Two-block CNN
Block 2 is the studied block.
Block 2 is a standard bottleneck.
It compresses the channel space: 8 → 4 → 8.

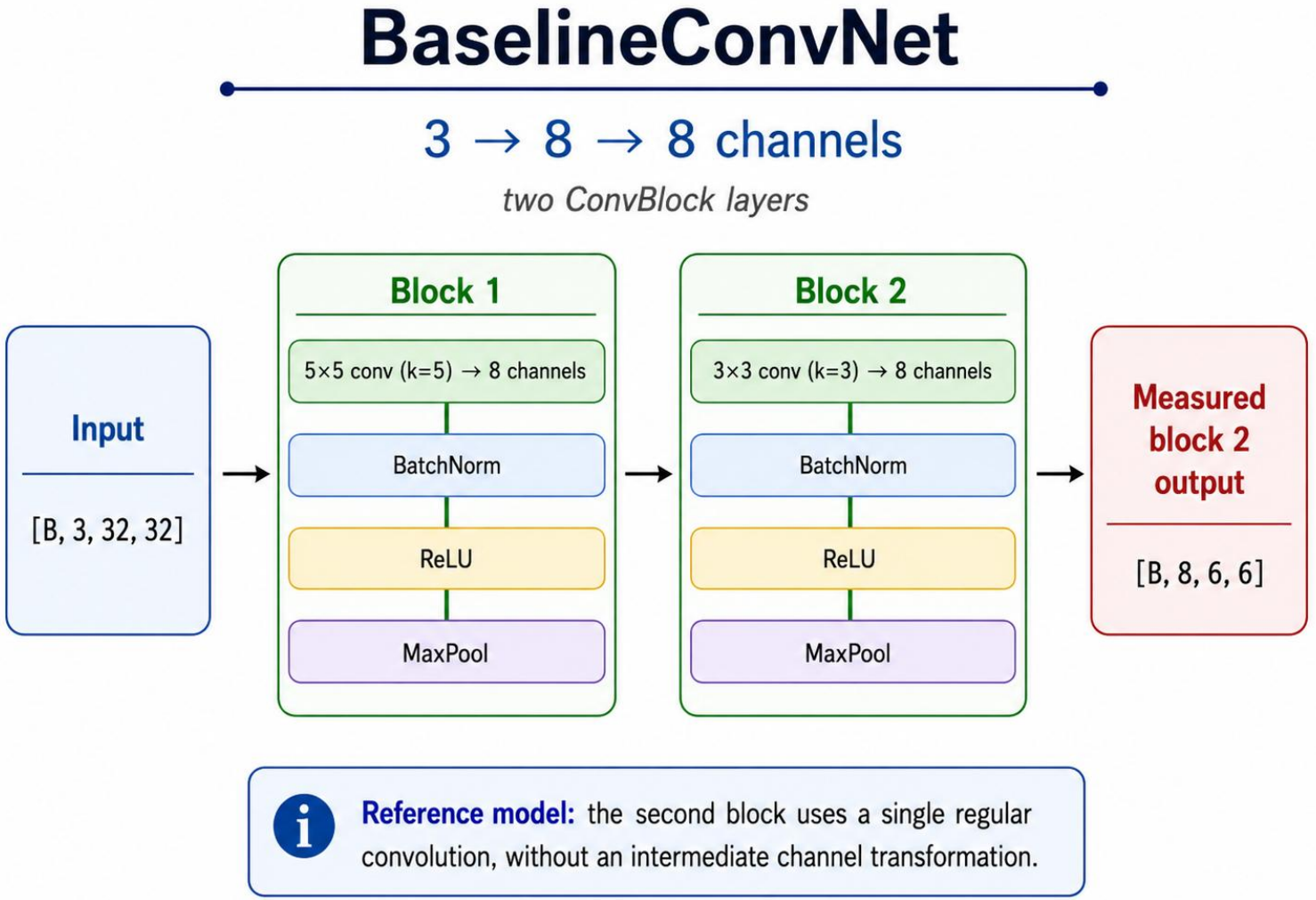
InvertedBottleneck



Two-block CNN
Block 2 is the studied block.
Block 2 is an inverted bottleneck.
It expands the channel space: 8 → 16 → 8.

Channel-matched comparison: all models are measured at the output of **block 2 with C = 8 channels**. They are not perfectly matched in **spatial resolution or parameter count, but close enough to be comparable in scale.**

Compared architectures: BaselineConvNet



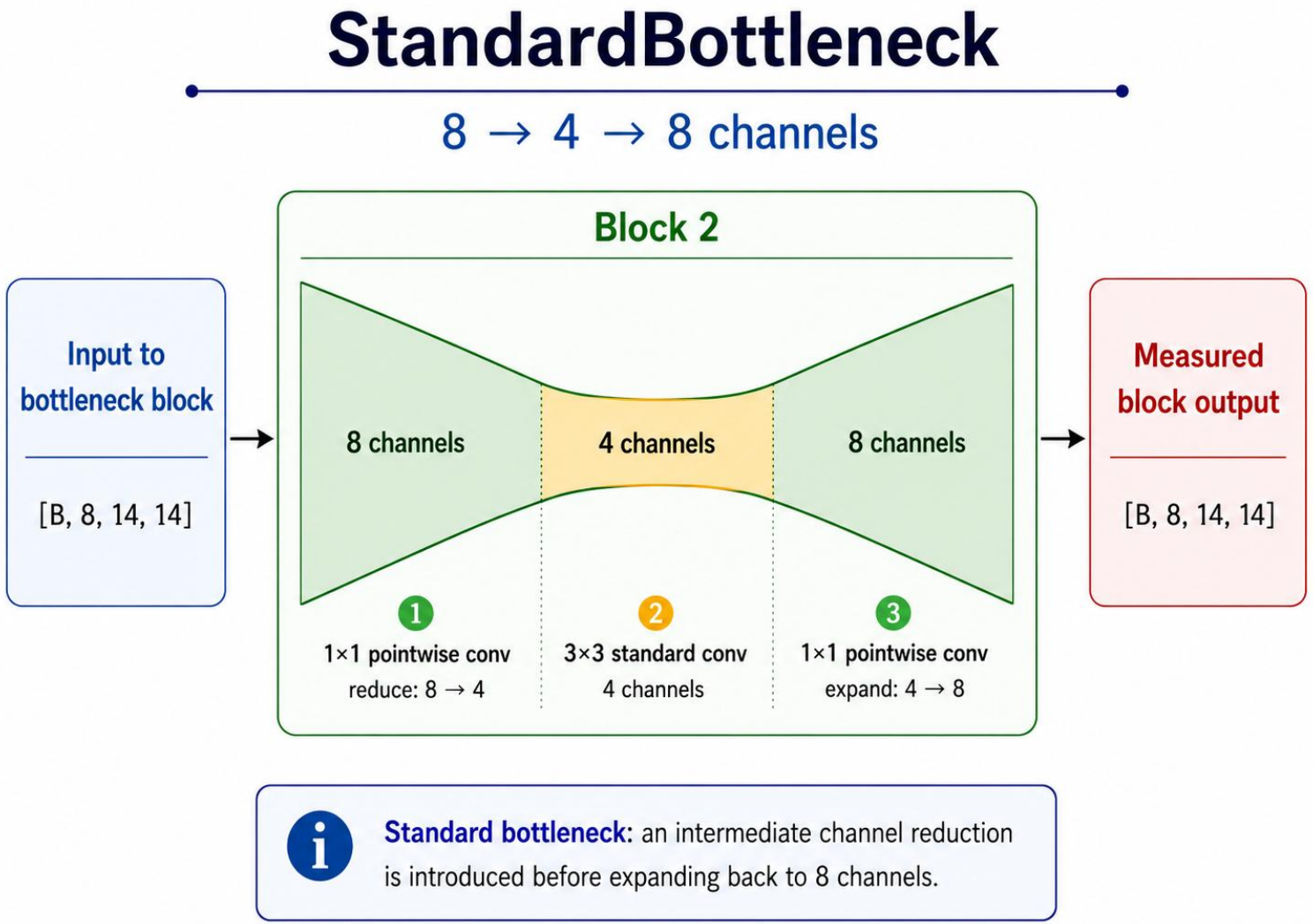
Two regular convolutional blocks.

No intermediate dimensional transformation.

3 → 8 → 8 channels.

Reference model to see what bottlenecks induce and what naturally occurs in CNNs.

Compared architectures: Standard bottleneck

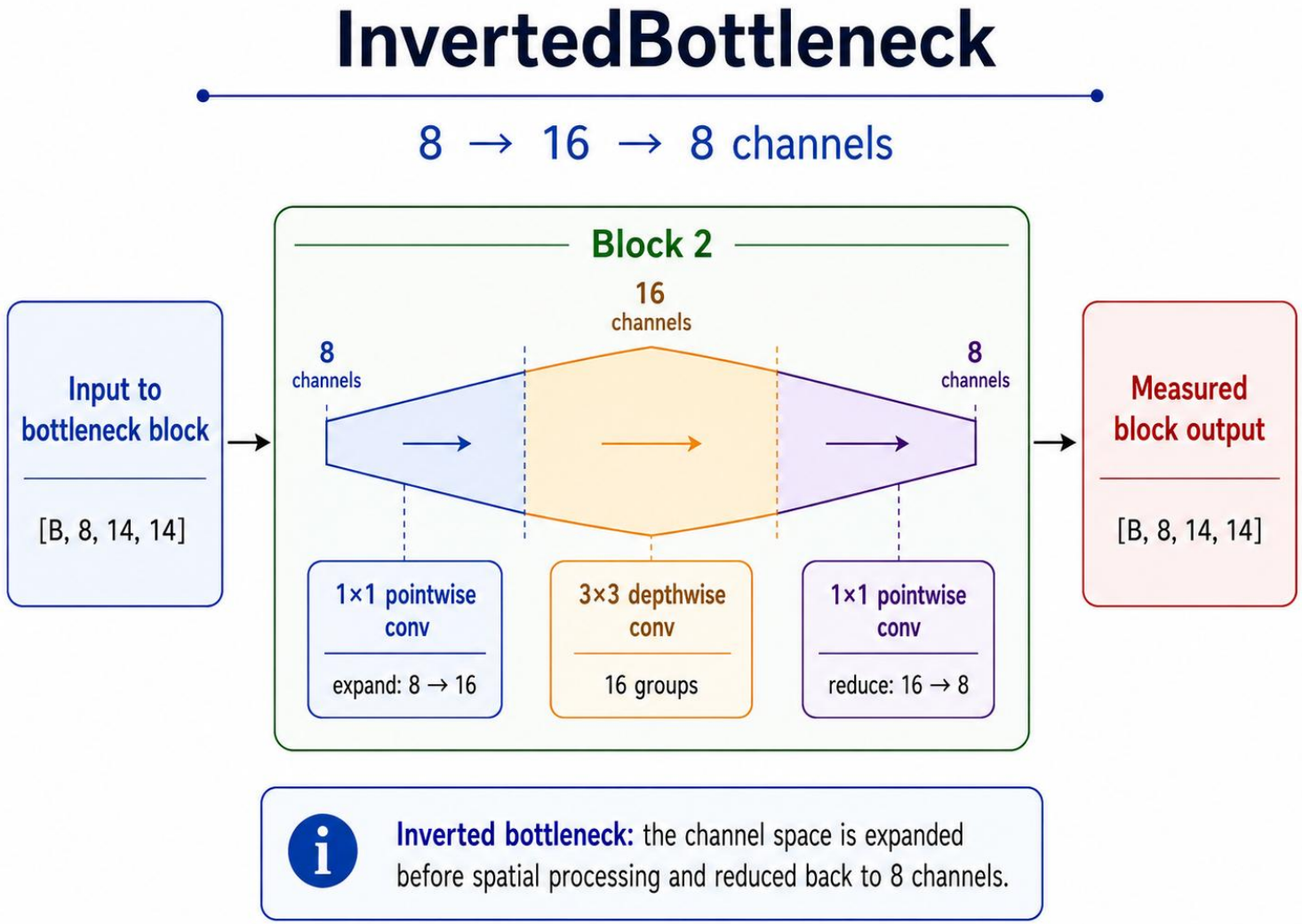


8 channels down to 4, then back to 8.

Internal compression in the channel space.

Useful to study what happens when the representation is reduced before being expanded again.

Compared architectures: Inverted bottleneck



8 channels up to 16, then back to 8.

Internal expansion before spatial processing.

Useful to compare how expansion and compression affect feature organization differently.

Dataset and training protocol

Dataset

- Four-class subset of **CIFAR-10**.
- **20,000 training images** and **4,000 test images**.
- Standard **32 × 32 RGB inputs**.
- The subset keeps the task adapted to the small two-block models studied here.

4 classes · 20,000 train · 4,000 test · 32×32 RGB

Training protocol

- Same conditions for all architectures: loss function, optimizer, batch size, and number of epochs.
- Images normalized using CIFAR-10 mean and standard deviation..
- The models reached similar validation accuracies, around **67–70%**.
- Models learned the task to a comparable level, but accuracy is **not the main focus**.

Representation measures :

Cosine similarity

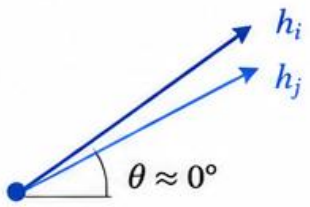
local
same behavior in specific inputs

pairwise
compares channels two by two

channel alignment
measures direction similarity

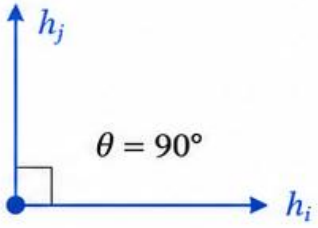
proxy for overlap
high similarity suggests overlapping activations

Aligned (high cosine similarity)



$\cos(\theta) \to 1$
very similar directions

Orthogonal (low cosine similarity)



$\cos(\theta) \to 0$
independent directions

What it tells us: How similar the channels are to each other
➡ local, pairwise overlap

Effective rank

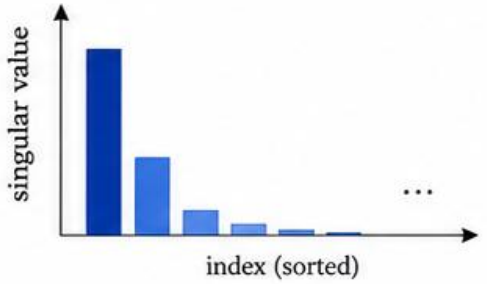
global
considers all channels together

effective dimensions
estimates how many directions are used

low = concentrated
information packed in few directions

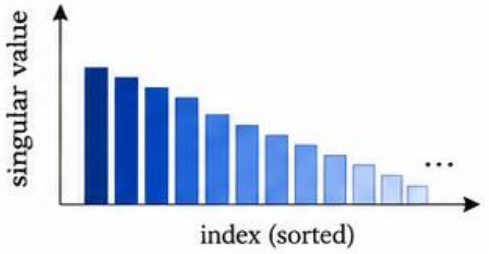
high = distributed
information spread across many directions

Low effective rank (concentrated)



few dominant directions
representation is concentrated

High effective rank (distributed)

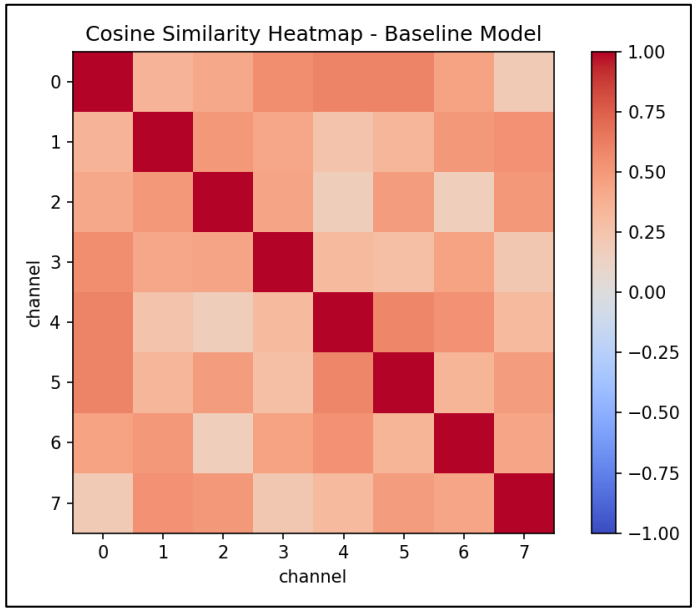


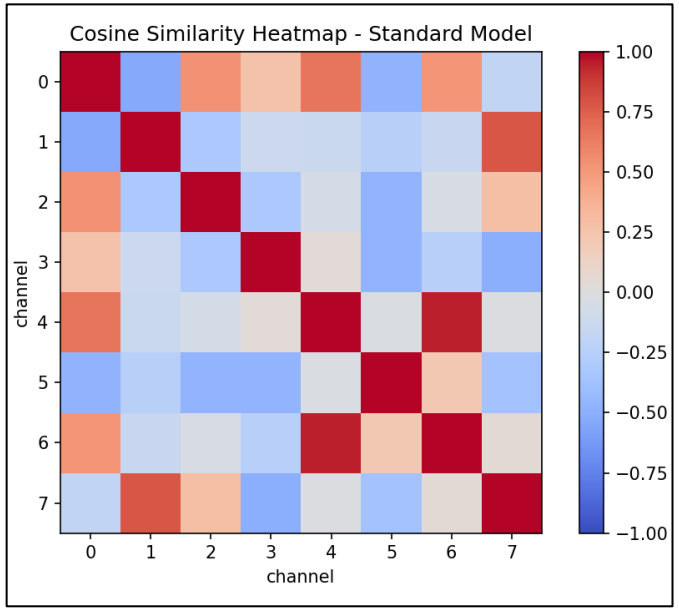
many directions contribute
representation is distributed

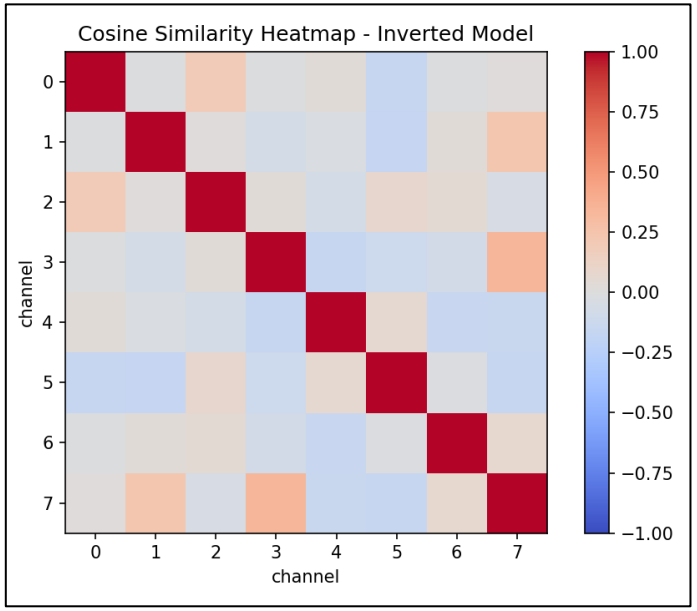
What it tells us: How many independent directions are effectively used
➡ global, overall concentration vs. distribution

These measures remain indirect proxies. They show whether representations overlap, but they do not prove superposition, because they do not tell us which actual features are being shared.

Results: cosine similarity







Interpretation

- The baseline already shows relatively high cosine similarity.
- -> So channel overlap can occur even without an explicit bottleneck.
- The standard bottleneck shows the highest maximum cosine similarity.
- -> Some channel pairs become strongly aligned after compression-expansion.
- The inverted bottleneck shows the lowest cosine similarity.
- -> Less pairwise alignment after the expansion.

Overlap is not specific to bottleneck architectures.

Results: quantitative interpretation

Cosine similarity statistics

Architecture	Mean abs offdiag	Max abs offdiag	Frac > 0.5
Baseline	0.4198	0.7063	0.2500
Standard BT	0.3247	**0.9199**	0.2500
Inverted BT	**0.1450**	0.4398	**0.0000**

Effective rank statistics

Architecture	Effective rank	Normalized effective rank
Baseline	7.0406 / 8	0.8801 = 88%
Standard BT	**4.1000 / 8**	**0.5125 = 51%**
Inverted BT	**7.7319 / 8**	**0.9665 = 97%**

Baseline

Cosine: relatively high overlap
Overlap can occur without bottleneck
Rank: high , close to max
Not globally concentrated
Superposition: overlap alone ≠ proof

Standard bottleneck

Cosine: highest max similarity
Strongly aligned channel pairs
Compression: 8 → 4 → 8
Rank: low , 4 /8
Strongest superposition-like pattern
Superposition: shared directions possible

Inverted bottleneck

Cosine: lowest similarity
Less pairwise alignment
Less channel overlap
Rank: very high , 7.7 /8
Distributed representation
Superposition : Less compatible with concentration

Conclusion and open perspective

Conclusion

The main point is that these results are suggestive, but not definitive.

Cosine similarity and effective rank help describe the geometry of the representations.

They can suggest overlap, but they cannot prove superposition on their own.

They can't tell us which features are encoded by the channels.

Open Perspective

Sparse Autoencoders could help identify the actual features inside the activations.

Check whether the observed overlap means distinct features sharing the same representational directions,

or simply general channel correlation.

Thank you for listening